

Bihar Engineering University, Patna**B.Tech. 1st Semester Examination, 2023****Course: B. Tech.****Time: 03 Hours****Code: 100104****Subject: Programming for Problem Solving****Full Marks: 70****Instructions:-**

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):**[2 × 7 = 14]**

- (a) What is the output of the following code?
- ```
int x = 10;
x++;
printf("%d", x);
```
- (i) Error (ii) 10 (iii) 11 (iv) 9
- (b) Which function is used to dynamically allocate memory in C?
- (i) malloc() (ii) calloc() (iii) realloc() (iv) all of the above
- (c) What is the output of the following code?
- ```
#include<stdio.h>
int main() {
    char *p = 0;
    *p = 'a';
    printf("%c", *p);
    return 0;
}
```
- (i) It will print 'a' (ii) It will print 0 (iii) Compile time error (iv) Runtime error
- (d) What are the elements present in the array of the following C code?
- ```
int array[6] = {5};
```
- (i) 5, 5, 5, 5, 5, 5 (ii) 5, 0, 0, 0, 0, 0  
(iii) 6, 6, 6, 6, 6, 6 (iv) (garbage), (garbage), (garbage), (garbage), (garbage), 5
- (e) What will be the output of the following C code?
- ```
void main(){
    int i = 0;
    while (i < 10){
        i++;
        printf("hi\n");
    }
    while (i < 8){
        i++;
        printf("hello\n");
    }
}
```
- (i) hi is printed 8 times, hello 7 times and then hi 2 times
(ii) hi is printed once, hello 7 times
(iii) hi is printed once, hello 7 times and then hi 2 times
(iv) None of the above.
- (f) What does the following declaration mean?
- ```
int (*ptr)[10];
```
- (i) ptr is array of pointers to 10 integers. (ii) ptr is a pointer to an array of 10 integers.  
(iii) ptr is an array of 10 integers. (iv) ptr is a pointer to array of 10 characters.
- (g) C is \_\_\_\_\_ type of programming language?
- (i) Object Oriented (ii) Procedural (iii) Bit level language (iv) Functional
- (h) Choose the right C statement.
- (i) int my\_age = 10; (ii) int my.age = 10; (iii) my age = 10; (iv) All are correct

- (i) Consider the following integer 2D array, which is stored in memory with the base address 1400. What would be the address of the element 14? [Assume that an integer value takes 4 bytes of memory space.]

`int a[3][5] = {{1,2,3,4,5}, {6,7,8,9,10}, {11,12,13,14,15}};`

- (i) 1440                      (ii) 1444                      (iii) 1448                      (iv) 1452

- (ii) Consider the following declaration of the variables:

`int x, A[5][6], *p = &A[0][0];`

Which of the following is/are the correct expression(s) to access `A[3][5]` and assign it to `x`?

- (i) `x = *(p+23);`      (ii) `x = *(p+28);`      (iii) `x = (*(A+5)+3);`      (iv) `x = (*(A+3)+5);`

- Q.2** (a) Explain the difference between auto and static type storage variables with suitable example codes. [7]  
(b) Differentiate between structures and unions in C. Write a C program to detect whether the computer is little endian or big endian using unions. [7]
- Q.3** (a) Write an efficient C program to search if the integer 24 is present in the given sorted array or not. If it is present, print the index where it is found. [7]  
(b) Explain the insertion sort algorithm and its steps using an example. Consider the array: [5, 2, 4, 6, 1, 3]. [7]
- Q.4** (a) Describe the concept of pointers in C programming language. Explain how pointers work and discuss their importance in programming, particularly in memory management and efficient data manipulation. [7]  
(b) Write a C program to copy contents of one file to another. While doing so replace all lowercase characters to their equivalent uppercase characters. [7]
- Q.5** (a) Write a C Program to find the Euclidian distance between two points P(x, y) and Q(a, b) using structure representation of points P and Q. [7]  
(b) Differentiate between call-by-value and call-by-reference with suitable examples. [7]
- Q.6** (a) Define arrays in C and explain their declaration and usage. Write a C program to reverse the content of an integer array without using auxiliary array. [7]  
(b) Define implicit and explicit type casting in C. Provide code examples illustrating when each type of casting occurs and their potential side effects. [7]
- Q.7** (a) Explain the concept of nested loops. Describe how the break and continue statements alter the flow of control within C loops. [7]  
(b) Explain the concept of algorithmic complexity and its significance in computer science. Using examples, illustrate how different algorithms exhibit various orders of complexity, such as constant time ( $O(1)$ ), linear time ( $O(n)$ ), logarithmic time ( $O(\log n)$ ), quadratic time ( $O(n^2)$ ), and exponential time ( $O(2^n)$ ). [7]
- Q.8** (a) The first 6 numbers of the Fibonacci series are: 0 1 1 2 3 5. Write a C program using recursion to find and print the Fibonacci series for first  $n$  terms. [7]  
(b) Write a C program using recursion to find factorial of a given number  $n$ . [7]
- Q.9** (a) Write a C program for the following:  
(i) Reverse a string without using `strrev()`.  
(ii) Concatenate two strings without using `strcat()`.  
(b) Write a C program to find result matrix after multiplying two given matrices using 2-D arrays. Check the multiplication criteria also while deriving the logic. [7]